Image Processing

Q.1) What is Digital Image Processing? Explain the application of DIP.

→ • Digital Image Processing (DIP) is the technique of improving or modifying an image using a computer so that useful information can be obtained from it.

• In DIP, an image is treated as a digital signal (Pixel) & various operations like enhancement, restoration, compression, & analysis are applied.

• The most common applications of digital image processing are

(I) Medical field - Xray, MRI, CT San Analysis
(II) Satellite imaging - Weather forcasting & remote sensing
(III) Traffic control - Vehicle & number plate detection
(IV) Biometrics - Identify verification
(V) Entertainment - Photo editing & video processing

Q2) Describe the Fundamental Steps in Digital Img Processing

→ (I) Image Acquisition : Image is captured using a camera or sensor & converted into digital form.

(II) Image Enhancement : Image quality is improved by adjusting brightness, contrast, or sharpness.

(III) Image Restoration : Noise & blur are removed using mathematical methods.

(IV) Color Image Processing : Color information is used to analyze & enhance images.

(V) Image Compression : Image size is reduced for easy storage & fast transmission.

(VI) Morphological Processing : Shape & Structure information of objects is extracted

(VII) Image Segmentation : Images is divided into meaningful regions or objects.

Q.3) Brief about human eye structure.

→ The human eye works like a camera & acts as an image sensing device
- Main Parts of the human eye are :

(i) Cornea - Transparent front part that allows light to enter the eye.

(ii) Iris - Controls the amount of light entering the eye.

(iii) Pupil - small opening through which light enters.

(iv) lens - focuses the light onto the retina.

(v) Retina - Inner layer where image is formed & converted into Signals.

(vi) Rods - Help in vision under low light (black & white vision)

(vii) optic nerve - Carries signal from retina to the brain.

Date
Page

Q.5) Write a short note on Image Sampling & Quantization.

→ Image Sampling & Quantization are the two basic steps used to convert an analog image into a digital image.

Image Sampling:

- image Sampling is the process of converting a continuous image into discrete spatial coordinates (pixels)
- It decides the spatial resolution of the image.

More sampling → more pixels → better image quality
Less sampling → loss of image details

Image Quantization

- Image quantization is the process of converting continuous intensity values into finite gray levels.
  It decides the intensity resolution of the image.

More gray levels → smoother image
fewer gray levels → distortion & false contouring

Q) What is filtering? What do you understand by the term spatial filtering?

→ Filtering is a process used in image processing to modify pixel values to achieve tasks like; noise removal, smoothing & edge enhancement.

- It works by applying a filter/mask to the image to change pixel values.

☆ Spatial Filtering

- Spatial filtering means applying a filter directly on image pixels in the spatial domain.

- General Formula:

$$g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) \cdot f(x+s, y+t)$$

where

$f(x,y)$ = input image

$g(x,y)$ = output image

$w(s,t)$ = filter mask

$(2a+1) \times (2b+1)$ = mask size (e.g., 3X3)

Input Image   Mask

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} \longleftrightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

output pixel = average of all 9 pixels.

Date
Page

Q.) Write a short note on smoothing filters.

→ Smoothing filters are lowpass filters used to reduce noise & make the image look softer by decreasing intensity changes.

— How They Work:

- These filters replace each pixel with a value based on its neighbouring pixels, usually the average.
- This removes small unwanted details & produces a smooth image.

— Median Filter (Non-Linear Smoothing)

- Another imp smoothing filter is the median filter, where the central pixel is replaced by the median of neighbouring pixels.

— Uses of Smoothing Filters:

- Noise reduction
- Smoothing satellite or medical images
- Blurring sensitive regions
- Reducing small background patterns
- Cleaning binary images

Date
Page

Q) Write a short note on Sharpening filters.

→ • Sharpening filters are high-pass spatial filters used to highlight edges, boundaries, & fine details in an image.
• They work by enchancing regions where intensity changes quickly, which typically represent edges or imp features.
• Sharpening is commonly used to improve image clarity or to restore details lost due to smoothing.

→ How Sharpening Filters Work:-

- Sharpening filters increase the contrast between a pixel & its neighbours.
- Unlike smoothing (which reduces differences), sharpening amplifies differences.

→ Purpose of Sharpening:

• Edge enhancement
• Highligthing object boundaries
• Making images look clearer
• Extracting fine details
• Restoring blurred
• Improving visual quality of photographs or medical images.

Q.) Explain the model of image degradation & reconstruction with a diagram?

→ The image degradation & reconstruction model explain how an original image becomes blurred or noisy & how we try to restore it back using mathematical filters.

- Image Degradation Model

- $f(x,y)$ represents the original image
- The image passes through a degradation system
- $h(x,y)$ represent the blur function
- $\eta(x,y)$ represent noise added during transmission
- Due to blur & noise the output becomes a degraded image $g(x,y)$

- Image Reconstruction (Restoration) Model

- $g(x,y)$ (degraded image) is given as input to a restoration filter
- $R(x,y)$ represents the restoration filter (Inverse filter)
- The aim of restoration is to remove blur & noise as much as possible.
- The output is the restored image.

Module -2

Q.) Explain the color Image Compression.

→ • Color Image compression is the process of reducing the size of a color image by removing unnecessary or repeated information while keeping the visual quality acceptable

• Since color images contain three components (R, G, B) they require more storage than grayscale image, so compression becomes essential.

→ Steps in Color Image Compression:

(I) Convert the image from RGB to YCbCr
(II) Downsample the color components (Cb & Cr)
(III) Apply a transform (like DCT in JPEG)
(IV) Quantize the transform values
(V) Encode the quantized values
(VI) Apply entropy coding (Huffman / Arithmetic)
(VII) Get the compressed image

→ Types of Color Image Compression:

(I) Lossless Compression	(II) Lossy Compression
• No loss in image quality	• Slight quality loss but much higher compression ratio
• Used for medical img, technical drawings	• Used in photos, video
• Eg: LZW, PNG, GIF, Huffman	• Eg: JPEG, WebP MP3, MP4
• Used when accuracy is imp.	

Date
Page

Q8.) What is data compression?

→ • Data compression is the process of reducing the size of data so that it takes less storage space & can be transmitted faster
• It removes unnecessary or repeated information while keeping the imp. information intact.

→ Types of Data Compression:

→ How Compression Works:

• Removes repeated patterns → dictionary coding (LZW)
• Removes unnecessary bits → entropy coding (Huffman/Arithmetic)
• Removes less important img details → transform coding (DCT/JPEG)

Q) Explain the color fundamentals.

→ • Color fundamentals deal with the basic concepts of color, how human perceive color, & how color information is represented in digital images.
• Understanding color fundamentals is imp for color img processing, compression & enhancement.

→ Properties of Color:
• Color is described using three basic properties -

a) Hue - Represent the actual color (red, green, blue etc)
b) Saturation - Represent the purity or richness of a color
c) Intensity - Represent the amount of light.

→ Human Color Perception:

The human eye can distinguish colors due to special cells in the retina called cones.

There are three types of cones:

Red-sensitive cones	Green sensitive cones
Blue sens. cones	

→ Color Models:
color models are mathematical ways to represent colors -

• RGB - Used in displays & Cameras
• CMY / CMYK - used in printers
• HSV / HSI - used in image processing
• YCb Cr - Used in compression & video systems.

Q.) Explain the Arithmetic Coding with an example.

→ • Arithmetic Coding is a lossless data compression technique that represents an entire message as a single decimal number between 0 & 1.

• Instead of assigning fixed bit codes to each symbol (like huffman) arithmetic coding encodes the whole message in one interval.

Date
Page

Q.) Write a short note on Erosion & Dilation

→ • Erosion & Dilation are the two basic operations in morphological image processing.
• They are mainly used on binary images & are based on the shape of a small matrix called the structuring element (SE)

(A) Erosion:

erosion shrinks or thins the objects in an image. It removes pixels from the boundaries of objects.

Diagram

– Effect:
• object become smaller • small noise is removed

(B) Dilation

Dilation grows or expands the objects in an image. It adds pixels to the boundaries of objects.

D

– Effect:
• Object become larger • Fills small holes

– Uses of Erosion & Dilation

- Noise removal
- Filling small gaps
- Separating or connecting objects
- Preparing image for segmentation
- Shape analysis

Date
Page

Q.) Write a short on Opening & closing.

→ • Opening & Closing are morphological operations derived from erosion & dilation.
• They are used to remove noise, smooth object boundaries & improve object shapes in binary images.

(A) Opening – is the process of erosion followed by dilation using the same structuring element.

$$\text{Opening} = (A \ominus B) \oplus B$$

Effect:
- Removes small objects & noise
- Breaks thin connection
- smooths objects boundaries

(B) Closing – is the process of dilation followed by erosion using the same structuring element.

$$\text{Closing} = (A \oplus B) \ominus B$$

Q) What is thresholding? Explain global Thresholding.

→ Thresholding is an image segmentation technique used to seprate objects from the background based on pixel intensity values.

Suppose;
- we choose a threshold value (T)
  - pixels above T → object
  - pixels below T → background

After thresholding, the image usually becomes a binary image (black & white)

→ Global Thresholding is a method where one single threshold value (T) is used for the entire image.

Steps;
- choose a threshold value T
- compare every pixel with T
- If pixel ≥ T → assign white (object)
- If pixel < T → assign black (background)

— Example:
Assume threshold T = 100

pixel value	Result
120	object (white)
90	Background (black)
150	Object (white)
60	Background (black)

So the img is separated into object & background.